

Gazebo: Let Your Robots Do The Talking

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Behind every robot there is a story of how it was made. Though there was no mama robot or papa robot, there is a story of ideation, prototyping, debugging and the usual repeat till satisfied. In my humble opinion, if humans could be brought into this world the same way, we would all be far better off, but that is a different story and a different scheme of thinking.

A robotics simulator is kind of a nanny equivalent for a robot baby. It teaches it the basics of obeying commands, lets it foster in a safe environment, not unlike how Neo trained with Morpheus in the movie Matrix. Robots can learn from this environment and proceed to become the best version of the idea that birthed them. There are a lot of solutions easily available for people interested in taking up this task but, without doubt, the one solution which I am a huge fan of is Gazebo simulator.

Development for Gazebo began in the fall of 2002 at University of Southern California, USA. To fill the need for simulating robots in outdoor conditions, bringing together the various elements that affect the body of one, Dr Andrew Howard and Nate Koenig started work on a high-fidelity simulator. The name Gazebo, which means an outdoor pavilion-like structure, was to reflect the fact that the work was to simulate outdoor environments. A little too easy, but the fact remains that most of the work on Gazebo is done to simulate indoor environments.



Fig. 1: Drone on Gazebo



Fig. 2: Mars rover on Gazebo

The simulator to have

An ideal robotics simulator makes testing of environments and the actual robot very simple. It makes it possible to rapidly test algorithms, design robots, perform regression testing and train artificial intelligence (AI) systems that can be adaptive as well as possessive of

System requirements

Currently, Gazebo is best used on Ubuntu, the Linux flavour. You will need a computer that has:

- A dedicated GPU
- Intel i5 or higher
- 500MB of disk space
- Ubuntu trusty, or later (recommended)